DRFL-VAT: Deep Representative Feature Learning With Virtual Adversarial Training for Semisupervised Classification of Hyperspectral Image

Jialong Chen, Yuebin Wang, *Member, IEEE*, Liqiang Zhang, *Member, IEEE*, Meiling Liu, and Antonio Plaza, *Fellow, IEEE*

***Abstract*— While deep learning algorithms have achieved good results in hyperspectral image (HSI) classification, several supervised classification algorithms rely on a large number of labeled samples to get adequate performance. Collecting a large number of labeled samples is expensive in many real applications. To address this issue, a novel semisupervised HSI classification framework called deep representative feature learning (DRFL) with virtual adversarial training (DRFL-VAT) is developed in this article. By embedding the local manifold learning (LML) into the fully connected layers of a convolutional neural network (CNN), our newly developed DRFL can learn representative features. The VAT regularization is adopted to exploit the prediction label distribution of training samples and addresses the overfitting problem. Finally, the objective function of DRFL-VAT is solved by a customized algorithm. We test our method on three widely public HSI datasets and our results show that our method is competitive when compared to other state-of-the-art approaches.**

***Index Terms*— Convolutional neural network (CNN), hyperspectral image (HSI) classification, local manifold learning (LML), semisupervised learning, virtual adversarial training (VAT).**

I. Introduction

HYPERSPECTRAL images (HSIs) comprise hundreds of spectral bands ranging from the visible to infrared parts of the electromagnetic spectrum and contain rich spatial and spectral information which is very useful in many applications [1]–[5]. HSI classification is also an active area. The large spatial variability of spectral signatures and the limited availability of training data (compared to the high dimensionality of HSIs) are two major challenges in this context [6].

Lots of HSI classification methods have been developed to address these problems, and these works according to the use condition of labeled information can be divided into three categories: supervised, semisupervised, and unsupervised classification methods. Traditional maximum likelihood classification is a typical supervised statistical classification method [7]. The support vector machine (SVM), as a supervised kernel classifier, separates unknown samples by finding an optimal decision hyperplane [8]. Compared with the classification methods based on spectral information alone, the fusion of spectral and spatial information in HSI analysis improves the discrimination ability of classifiers in complex scenes [9]. Markov random fields are also often utilized to explore spatial-context information and enhance the performance of classifiers [10]. Conditional random fields (CRFs) have been adopted to predict the class of each superpixel, obtained by compressing HSI datasets [11]. Thanks to recent advances in deep learning for computer vision, many supervised deep learning methods also have been proposed to classify HSIs by using spectral information and spatial information. In [12], a spectral convolutional neural network (CNN) is directly used to classify HSIs in the spectral domain. Spatial–spectral CNNs can explore simultaneously spectral and spatial information in the HSI for classification tasks and achieve a better classification performance than spectral-based CNN models [13], [14]. For supervised classification methods, a large number of labeled samples is generally needed to obtain high classification accuracy, while the production of numerous labeled samples is time-consuming.

Unsupervised feature learning can effectively extract the feature representations of samples in HSI without labeled information, and the feature representations learned are usually used for classification or clustering tasks. Conventional linear feature learning methods include principal component analysis [15], linear discriminant analysis [16], independent

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component analysis [17], and so on. Several nonlinear feature learning methods also perform well in feature extraction tasks for the nonlinear data [18]–[20], such as HSI data. Meanwhile, many deep unsupervised learning models have been employed for HSI feature extraction successfully. The stacked autoencoder (SAE) and deep beliefs network (DBN) are widely used to extract nonlinear feature representations in an unsupervised manner [21], [22]. Although several clustering techniques are proposed for HSI clustering tasks, often there are no available labeled samples to guide the process of feature learning, resulting in a poor clustering performance [23]. For the HSI classification tasks, the unsupervised feature learning methods are employed to extract feature representations which are fed into a simple classifier to classify HSI [24].

Semisupervised learning methods (which can simultaneously utilize the information of labeled samples and the structural characteristics of both labeled and unlabeled samples) are widely used for HSI classification. The SAE has been used to learn deep representative features for unsupervised classification of HSIs [25]. Generative adversarial networks (GANs) that contain a generative network and a discriminative network are used to classify HSIs by training a discriminator [26]. Subspace learning with CRFs uses the 3-D convolutional autoencoder to extract features and a CRF framework is adopted to improve the performance of subspace learning [27]. For the methods in [25]–[27], their classification framework can be divided into two steps: pretrained the feature extraction model on unlabeled samples, and supervised finetuning on labeled samples by using a simple classifier. A semisupervised classification method, called semisupervised deep learning using pseudo labels, utilizes the pseudo label information of unlabeled samples and the label information of labeled samples for HSI classification [28]. Based on the ladder network, a semisupervised CNN (SSCNN) has been used to directly extract spatial–spectral features for HSI classification [29]. The semisupervised variational GAN adopts an encoder–decoder network to learn the relationship between the latent space and the real image space in the context of HSI classification [30]. Graph learning can use labeled and unlabeled samples to construct a graph that reflects the relationship among samples [31]. The graph is constructed for semisupervised learning by local linear embedding, and the discriminative spatial information is employed to improve the classification results [32]. Superpixel feature vectors, which are obtained by considering the spectral covariance matrix, are employed to build a superpixel graph and a semisupervised framework is developed to classify HSIs in [33]. The probability matrix obtained by sparse representation is integrated into the sparse representation graph to get the discriminant sparse representation graph for semisupervised learning in [34].

However, the construction of a graph in graph-based semisupervised learning algorithms directly utilizes a distance function to calculate the relationships between the training samples, while the relationships may be ambiguous for the samples in the original nonlinear high-dimensional space [35]. In high-dimensional space, the two samples that are considered highly likely to belong to the same category because they are close to each other, sometimes do not have the same label information. Similarly, if a graph is directly created in local manifold learning (LML) by the samples in the nonlinear high-dimensional space, the relationships among samples are also ambiguously expressed. The local structure of data samples and distribution cannot be properly estimated with their ambiguous relationships. If two samples are compact in terms of the intrinsic geometry of the data distribution, their feature representation in a new space should keep a similar relationship [36]. Therefore, the feature representations of samples are considered for the construction of a graph in the LML. When the feature representations of samples are optimized, the relationships between the samples are gradually clear. Moreover, the construction of a graph usually based on all labeled and unlabeled samples and the label prediction of newly added samples increases extra computation cost because the graph needs to be built again.

Under the above considerations, a novel semisupervised learning framework called deep representative feature learning (DRFL) with virtual adversarial training (DRFL-VAT) is developed in this article for HSI classification purposes. The DRFL-VAT can learn highly representative features compared with other approaches. While the local smoothness of sample features is improved by the LML, the data label distribution is well characterized by the VAT regularization, which also can address the overfitting problem caused by the availability of training samples. In this work, CNN is first used to extract the basic spatial-spectral feature in the HSI by using a multiscale filter bank. By embedding the LML into fully connected CNN layers, a graph (which includes the relationships among samples) is built to exploit the local geometry structure of the manifold. The VAT can reduce the distance between the prediction label distribution of pure input data and the prediction label distribution of input data with adversarial perturbation and encourage the adopted model to output the same label distribution for data points from the same class. Both labeled and unlabeled samples are jointly used to adjust the whole model parameters in a semisupervised manner, so that the DRFL-VAT effectively alleviates the limitations introduced by the limited availability of labeled samples. Then, the objective function of DRFL-VAT is solved by a customized algorithm. During the training process, the feature representativeness of training samples is enhanced and the relationship among samples is more clearly expressed. An overview of our newly proposed DRFL-VAT for HSI classification is given in Fig. 1. We test our method on three widely used HSI datasets, and our experimental results show that the proposed method is competitive when compared with other state-of-the-art methods. The main contributions of our method can be summarized as follows.

1) To get more representative features for HSI classification, a new DRFL-VAT is developed to extract the features of both labeled and unlabeled samples.
2) To enhance the local smoothness of sample features and alleviate the requirement of a high number of labeled samples, LML and VAT are employed to explore the local properties of both labeled and unlabeled samples, and both labeled and unlabeled samples are used to

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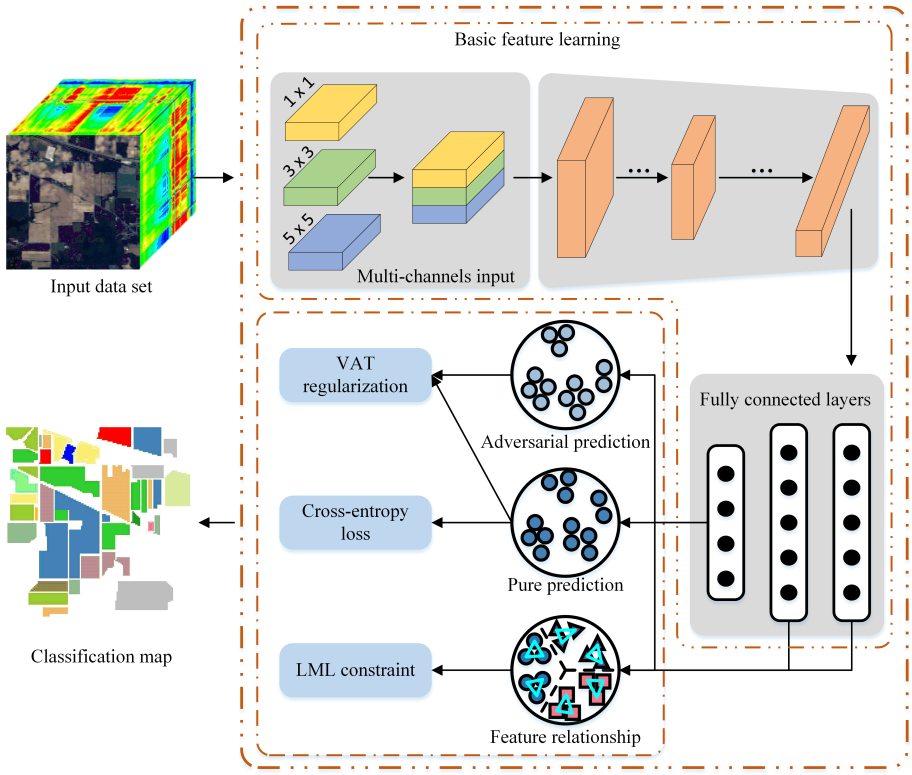


Fig. 1. Graphical illustration of our new DRFL-VAT framework for HSI classification. VAT regularization and LML constraint are utilized to extract features of both labeled and unlabeled samples. The objective function of DRFL-VAT, including the cross-entropy loss, VAT regularization, and LML constraint, optimizes the model parameters in a semisupervised manner.

optimize the model parameters in a semisupervised manner.

3) A customized algorithm is developed to optimize the DRFL-VAT. Our experiments show that the newly proposed method is very effective for HSI classification purposes.

II. Proposed Method

In this section, a novel semisupervised classification framework is proposed which is based on CNN, LML, and VAT. The motivation of our work is given in Section II-A. The basic feature learning procedure of HSI is described in Section II-B. Then, the constructions of LML and VAT are discussed in Section II-C. Finally, the objective function of DRFL-VAT is summarized in Section II-D.

A. Motivation

Supervised HSI classification methods usually rely heavily on a large number of labeled data to obtain high classification performance, while they also require an enormous amount of labor. Therefore, it is important to develop semisupervised classification methods that can ease the problem of limited labeled samples. The conventional CNN has been widely employed to extract features for HSI classification tasks. In this article, a CNN architecture that extracts spatial and spectral features simultaneously is introduced. For HSI classification, the representative features of samples can effectively explore the attribute characteristics of samples, and the relationship among samples can be clearly expressed by such representative features. To utilize the geometrical structure of the sample manifold to enhance the representativeness of features, the LML is embedded into each fully connected layer of the CNN. When the objective loss function of the model is evaluated by using a limited training dataset, the overfitting problem may appear. To alleviate such problems, VAT also is added to the final objective function to improve the model generalization performance. As a result, a new semisupervised framework called DRFL-VAT, which integrates the CNN, LML, and VAT, is developed in this work, and a custom algorithm is used to optimize the proposed method.

B. Basic Feature Learning

Since sparse connectivity and shared weights can greatly decrease the number of parameters, CNN is widely used for HSI classification [37], [38]. The basic structure of CNNs comprises convolutional layers, pooling layers, nonlinear layers, and fully connected layers. Convolutional layers and pooling layers are usually arranged alternately to obtain abstract features, and fully connected layers are employed to integrate features at the end of the CNN architecture. Abstract features are obtained from the original input data through the convolutional layers, pooling layers, and nonlinear layers and are stacked into a column vector as the input of fully connected layers. For classification purposes, a classifier is usually adopted at the last layer of the fully connected layers. The activation function, the pooling function, and the structure

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parameters of the CNN jointly affect the model performance. In [39], it discussed how to find a tradeoff between network depth, filter size, and number at a constrained time cost. The results show the importance of network depth in terms of improving performance while extremely large depth decreases accuracy.

In this section, we describe how the basic feature learning from HSIs is conducted by our CNN and the detailed parameters of the network. Let us denote a 3-D HSI dataset by $\mathbf{A} \in \mathbb{R}^{d\times m\times n}$ and a label matrix by $\mathbf{Y} \in \mathbb{R}^{c\times(m\times n)}$, where d denotes the number of bands and $m \times n$ denotes the number of pixels in each band, and c is the number of classes of the HSI dataset. First, pixel $\mathbf{X}_{.i} \in \mathbb{R}^d$ itself and its surrounding 3×3, 5×5 neighboring pixels are cropped in the HSI dataset so that the HSI dataset can be split into different patches with varying sizes, including $1 \times 1 \times d$, $3 \times 3 \times d$, $5 \times 5 \times d$. A boundary mirroring strategy is employed to solve the lack of pixel information around the edge pixels in the HSI [19]. Then, the patches are grouped in batches of size M and sent to the first convolutional layer. This approach can effectively speed up the training process of the model. The first convolutional layer is a multiscale filter bank that can explore local spatial features and spectral correlation simultaneously. The multiscale filter bank is composed of different scale convolutional filters of size $1 \times 1 \times d$, $3 \times 3 \times d$, $5 \times 5 \times d$, and the dimensionality of each filter is k^c. The $1 \times 1 \times d$ convolutional filter can obtain spectral information of pixels, while the $3 \times 3 \times d$ and $5 \times 5 \times d$ filter are employed to extract local spatial information among pixels [40]. The sizes of the three feature maps obtained from the first convolutional layer are 1×1, 3×3, and 5×5, respectively, and then the three feature maps are transformed to nonlinear feature maps by the leaky rectified linear (LeakyReLU) activation function. The LeakyReLU function can alleviate the vanishing gradient and dead ReLU problem [41]. To merge the three feature maps into a joint feature map, a space of two-pixel width filled with zeros is padded around the 1×1 feature maps, and a space of one-pixel width filled with zeros is padded around the 3×3 feature maps. Then, k^c feature maps of size 5×5 can be obtained, where $k^{c1} = 3k^c$. The k^{c1} feature maps are sent to the first max pooling layer, and the output feature map $M \times l^{mp1} \times l^{mp1} \times k^{c1}$ is sent to the second convolutional layer with k^{c2} filters of $l^{c2} \times l^{c2} \times k^{c1}$. Again, after applying the LeakyReLU activation function, k^{c2} feature maps (generated by the second fully convolutional layer) are sent to the second max-pooling layer. Then the final output feature map is of dimensions $l^{mp2} \times l^{mp2} \times k^{c2}$ for each input sample. In order to prepare for the training process (conducted by the fully connected layers), the final feature map of each pixel is stacked into a column vector of size l^{fc0}, where $l^{fc0} = l^{mp2} \times l^{mp2} \times k^{c2}$, and the column vector is sent as input data to the first fully connected layer.

C. Description of the Representative Features Learning Process With DRFL-VAT

1) Summary of Basic Features by Multiple Fully Connected Layers: In the CNN, the fully connected layers are used to summarize the output features from previous convolutional layers. Our fully connected area consists of three fully connected layers, and every fully connected layer is constructed with a_1, a_2, and c neurons, respectively. Given a feature matrix $\mathbf{Z}^{(0)} \in \mathbb{R}^{l^{fc0}\times M}$, which denotes the input data of the fully connected layers, the output feature of each fully connected layer is calculated by the following expression:

$$\mathbf{Z}^{(u)} = \begin{cases} f\left(\mathbf{P}^{(u)}\right), & u = 1, 2 \\ \mathbf{P}^{(u)}, & u = 3 \end{cases} \tag{1}$$

where $u = 1, 2, 3$ represents uth the fully connected layer, $\mathbf{Z}^{(u)}$ is the output feature vector of the uth fully connected layer, and $f(\cdot)$ is an activation function such as LeakyReLU function. Here, we have

$$\mathbf{P}^{(u)} = \mathbf{W}^{(u)}\mathbf{Z}^{(u-1)} + \mathbf{B}^{(u)} \tag{2}$$

where $\mathbf{W}^{(u)}$ is the matrix of the uth fully connected layer, and $\mathbf{B}^{(u)}$ is the bias matrix of the uth fully connected layer.

Using the Softmax function for HSI classification, we can have the result as follows for $\mathbf{X}_{.j}$ (in a batch of input data):

$$\mathbf{O}_{.j} = \frac{\exp\left(\mathbf{Z}^{(3)}_{.j}\right)}{\sum_{i=1}^{c}\exp\left(\mathbf{Z}^{(3)}_{ij}\right)}. \tag{3}$$

The cross-entropy loss function can obtain the error between the prediction label distribution $\mathbf{O}_{.j}$ and the true label distribution $\mathbf{Y}_{.j}$ of the input data $\mathbf{X}_{.j}$. Every batch of the input data should contain the labeled and unlabeled samples to ensure that the adjustment of parameters for the whole model is always conducted in a semisupervised manner. The cross-entropy loss function is given by

$$\Phi_{\text{cel}} = -\frac{1}{M_0}\sum_{j=1}^{M_0}\sum_{k=1}^{c} 1\left\{k = \mathbf{Y}_{kj}\right\}\ln(\mathbf{O}_{kj}) \tag{4}$$

where M_0 is the number of labeled samples in a batch of the input data, and $1\{k = \mathbf{Y}_{kj}\}$ is 1 when the value of k equals the value of $\mathbf{Y}_{kj}$.

2) Representative Feature Learning With Local Constraints: Previous researches [40], [42], [43] have shown that the geometrical structure of data manifolds is important for representation learning and clustering. By constructing local constraints between neighboring samples, the LML can explore the local geometry of the samples manifold to improve the representativeness of features. With the loss of generality, a group of pixels of size M can be sent simultaneously to the neural network adopted to effectively speed up the training process. Here, the LML is applied to each fully connected layer of the CNN model. In each fully connected layer, a graph is constructed whose nodes correspond to the output features of the layer. The LML loss function Φ_{LML} is computed by the following expression:

$$\begin{aligned}\Phi_{\text{LML}} &= \sum_{u=1}^{3}\frac{1}{2}\sum_{m,n=1}^{M}\mathbf{V}^{(u)}_{mn}\left\|\mathbf{Z}^{(u)}_{.m} - \mathbf{Z}^{(u)}_{.n}\right\|_2^2 \\ &= \sum_{u=1}^{3}\text{tr}\left(\mathbf{Z}^{(u)}\mathbf{L}^{(u)}\left(\mathbf{Z}^{(u)}\right)^T\right)\end{aligned} \tag{5}$$

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where $\|\cdot\|_2$ is the l_2 norm, $\mathrm{tr}(\cdot)$ denotes the trace of a matrix, the Laplacian matrix $\mathbf{L}^{(u)} = \mathbf{D}^{(u)} - \mathbf{V}^{(u)}$, $\mathbf{D}^{(u)}$ is a diagonal matrix where the (l,l)th element is the sum of the lth row of $\mathbf{V}^{(u)}$, and $\mathbf{V}^{(u)}$ is the affinity matrix given by

$$\mathbf{V}_{mn}^{(u)} = \exp\left(-\frac{\left\|\mathbf{Z}_{\cdot m}^{(u)} - \mathbf{Z}_{\cdot n}^{(u)}\right\|_2^2}{2\omega^2}\right) \tag{6}$$

where ω is the heat kernel.

3) Representative Feature Learning With Prediction Label Distribution Estimation: Adversarial training exhibits good classification performance in many supervised classification tasks. Since labeled training samples are limited in our context, VAT is adopted to carry out several semisupervised tasks with limited labeled samples and numerous unlabeled samples [44]–[46]. VAT can obtain pseudo-label information for unlabeled samples by the adopted model, which can effectively develop adversarial training into the semisupervised learning field by considering output consistency. In the VAT, the adversarial data are obtained by adding adversarial noise into the pure data. The output distribution of the adversarial data should deviate as much as possible from that of the pure data. Then, the VAT encourages the model to obtain similar output distributions for the pure and adversarial data by optimizing the virtual adversarial loss. Given the pure data $\mathbf{X}_{\cdot i}$, the adversarial data $\mathbf{X}_{\cdot i} + \mathbf{T}_{\cdot i}^{\mathrm{adv}}$ and a true label $\mathbf{Y}_{\cdot i}$, the loss function of VAT is given by

$$\begin{cases} D\left[q(\mathbf{Y}_{\cdot i}|\mathbf{X}_{\cdot i}),\, p\left(\mathbf{Y}_{\cdot i}|\mathbf{X}_{\cdot i} + \mathbf{T}_{\cdot i}^{\mathrm{adv}}, \theta\right)\right] \\ \mathbf{T}_{\cdot i}^{\mathrm{adv}} = \underset{\mathbf{T};\|\mathbf{T}\|_2 \le \varepsilon}{\arg\max}\, D[q(\mathbf{Y}_{\cdot i}|\mathbf{X}_{\cdot i}),\, p(\mathbf{Y}_{\cdot i}|\mathbf{X}_{\cdot i} + \mathbf{T}, \theta)] \end{cases} \tag{7}$$

where $D(\cdot)$ is the Kullback–Leibler divergence, $\mathbf{T}_{\cdot i}^{\mathrm{adv}}$ is the adversarial perturbation of $\mathbf{X}_{\cdot i}$, and ε is used to limit the size of the adversarial perturbation. $p(\mathbf{Y}_{\cdot i}|\mathbf{X}_{\cdot i} + \mathbf{T}_{\cdot i}^{\mathrm{adv}}, \theta)$ is the prediction label distribution of $\mathbf{X}_{\cdot i} + \mathbf{T}_{\cdot i}^{\mathrm{adv}}$ and θ is the set of model parameters, including the weight matrix and the bias matrix. $q(\mathbf{Y}_{\cdot i}|\mathbf{X}_{\cdot i})$ is the true label distribution and it is unknown for unlabeled samples. Therefore, the current estimate $p(\mathbf{Y}_{\cdot i}|\mathbf{X}_{\cdot i}, \hat{\theta})$ is calculated to replace the label distribution of unlabeled samples. The loss function of the VAT is given by

$$\begin{cases} D\left[p(\mathbf{Y}_{\cdot i}|\mathbf{X}_{\cdot i}, \hat{\theta}),\, p\left(\mathbf{Y}_{\cdot i}|\mathbf{X}_{\cdot i} + \mathbf{T}_{\cdot i}^{\mathrm{adv}}, \theta\right)\right] \\ \mathbf{T}_{\cdot i}^{\mathrm{adv}} = \underset{\mathbf{T};\|\mathbf{T}\|_2 \le \varepsilon}{\arg\max}\, D[p(\mathbf{Y}_{\cdot i}|\mathbf{X}_{\cdot i}, \hat{\theta}),\, p(\mathbf{Y}_{\cdot i}|\mathbf{X}_{\cdot i} + \mathbf{T}, \theta)] \end{cases} \tag{8}$$

where $p(\mathbf{Y}_{\cdot i}|\mathbf{X}_{\cdot i}, \hat{\theta})$ is the prediction label distribution of $\mathbf{X}_{\cdot i}$, while $\hat{\theta}$ is the set of model parameters at a specific iteration step of the training process. For semisupervised learning, $p(\mathbf{Y}_{\cdot i}|\mathbf{X}_{\cdot i}, \hat{\theta})$ and the true label distribution $q(\mathbf{Y}_{\cdot i}|\mathbf{X}_{\cdot i})$ become close when the training process is sufficiently iterated. For a batch of the input data, the VAT loss function is defined as follows:

$$\Phi_{\mathrm{VAT}} = \frac{1}{M}\sum_{i=1}^{M} D\left[p(\mathbf{Y}_{\cdot i}|\mathbf{X}_{\cdot i}, \hat{\theta}),\, p\left(\mathbf{Y}_{\cdot i}|\mathbf{X}_{\cdot i} + \mathbf{T}_{\cdot i}^{\mathrm{adv}}, \theta\right)\right]. \tag{9}$$

In order to solve (9), $\mathbf{T}_{\cdot i}^{\mathrm{adv}}$ needs to be calculated. According to [38], $\mathbf{T}_{\cdot i}^{\mathrm{adv}}$ is obtained by

$$\mathbf{T}_{\cdot i}^{\mathrm{adv}} \approx \varepsilon \frac{\mathbf{g}_i}{\|\mathbf{g}_i\|_2}. \tag{10}$$

Algorithm 1 DRFL-VAT

Input: Training dataset $\mathbf{X}$, λ_1, λ_2, ε, ξ, ω, epoch, $e = 1$;
Procedure:
1: Grouping the patches of $\mathbf{X}$ in batches of size M
2: Initialize model parameters
3: **while** $e <$ epoch **do**
4: Get adversarial perturbation $\mathbf{T}^{adv}$ for the input data $\mathbf{X}$ using (10), (11);
5: Choose $\mathbf{X} + \mathbf{T}^{adv}$ as the input data and forward propagation using DRFL-VAT;
6: Choose $\mathbf{X}$ as the input data and forward propagation using DRFL-VAT;
7: Calculate total loss value using (12);
8: Backward propagation and update θ;
9: Update $e = e + 1$;

Then, $\mathbf{g}_i$ is given by

$$\mathbf{g}_i = \left.\frac{\partial D[p(\mathbf{Y}_{\cdot i}|\mathbf{X}_{\cdot i}, \hat{\theta}),\, p(\mathbf{Y}_{\cdot i}|\mathbf{X}_{\cdot i} + \mathbf{T}_{\cdot i}, \hat{\theta})]}{\partial \mathbf{T}_{\cdot i}}\right|_{\mathbf{T}_{\cdot i} = \xi \mathbf{d}_i} \tag{11}$$

where $\mathbf{d}_i$ is a randomly sampled unit vector and $\xi \neq 0$.

D. Objective Function of DRFL-VAT

The CNN, LML, and VAT are integrated into a unified framework (DRFL-VAT). In this section, we introduce the objective function of DRFL-VAT. Taking (4), (5), and (9) into account, the objective function of the DRFL-VAT can be summarized as follows:

$$\begin{aligned} \min_{\theta} \Theta(\theta) &= \min \Phi_{\mathrm{cel}} + \lambda_1 \Phi_{\mathrm{LML}} + \lambda_2 \Phi_{\mathrm{VAT}} \\ &= -\frac{1}{M_0}\sum_{j=1}^{M_0}\sum_{k=1}^{c} 1\{k = \mathbf{Y}_{kj}\} \ln\left(\mathbf{O}_{kj}\right) \\ &\quad + \lambda_1 \sum_{u=1}^{3} \mathrm{tr}\left(\mathbf{Z}^{(u)}\mathbf{L}^{(u)}\left(\mathbf{Z}^{(u)}\right)^T\right) \\ &\quad + \lambda_2 \frac{1}{M}\sum_{i=1}^{M} D\left[p\left(\mathbf{Y}_{\cdot i} \mid \mathbf{X}_{\cdot i}, \hat{\theta}\right),\, p\left(\mathbf{Y}_{\cdot i} \mid \mathbf{X}_{\cdot i} + \mathbf{T}_{\cdot i}^{\mathrm{adv}}, \theta\right)\right] \end{aligned} \tag{12}$$

where λ_1 and λ_2 are tradeoff parameters. A scheme of the training operation of the proposed method is summarized in Algorithm 1.

The whole model parameters are optimized according to (12), and only the parameters of $\mathbf{W}$ and $\mathbf{B}$ need to be optimized in our model by the gradient descent algorithm. The proper learning rate and the gradient of the objective function with respect to the network parameters make the network parameters change in the direction that promotes the convergence of the model. Proper initialization of network parameters can accelerate the convergence of the model loss value to a stable fluctuation range [47]. When the output features of the fully connected layers of the input samples and the parameters of $\mathbf{W}$ and $\mathbf{B}$ are simultaneously optimized, the relationships between the samples represented by the graph

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are also optimized. The mini-batch update strategy is selected to speed up the training process and the objective function is evaluated on the mini-batch of input samples. For the model training process, a batch of data, including labeled and unlabeled samples, is fed into the model in each training iteration. The cross-entropy loss Φ_{cel} based on the M_0 labeled samples in a batch of input samples calculates the distance between the output prediction distribution and the true label distribution. The LML loss and VAT loss, which are created by considering all samples in a batch of input samples, are used to enhance the smoothness of the output prediction distribution without extra-label information. In every training iteration, labeled and unlabeled samples are jointly employed to train the model parameters in order to ensure that the parameter optimization process is always performed in a semisupervised manner. The gradient calculation and parameter update in our model are accomplished by the existing open-source framework. The LML loss is created based on the output features of three fully connected layers and has a direct influence on the update of network parameters of three fully connected layers. To better understand the direct influence, the derivative computation of LML loss with respect to the parameters in each fully connected layer is listed in the Appendix.

III. Experimental Results

In this section, we evaluated the classification performance of the DRFL-VAT method on three public datasets. Then, we compared the classification results of DRFL-VAT with the related classification algorithms.

A. Experimental Datasets

Three benchmark HSI datasets are used to test the proposed algorithm.

1) The Indian Pines dataset, which was captured by the Airborne Visible Infra-Red Imaging Spectrometer (AVIRIS) sensor over the Indian Pine test site of North-western Indiana in 1992 contains 220 spectral reflectance bands with 145×145 pixels in the wavelength range of 0.4–2.5 μm, and the spatial resolution is 20 m. The bands covering the water absorption region (104–108, 150–163, 200) are discarded, retaining only 200 spectral channels. The dataset contains 16 classes and 10 249 labeled pixels.
2) The Salinas dataset, which was also captured by the AVIRIS sensor over Salinas Valley, California. It comprises 224 bands with 512×217 pixels with high spatial resolution (3.7-m pixels). The number of bands is reduced to 204 after removing 20 water absorption bands (108–112, 154–167, 224). This dataset contains 16 classes and 54 129 labeled pixels.
3) The University of Pavia (PaviaU) dataset, which was collected by the Reflective Optics Spectrographic Imaging System (ROSIS) sensor over the city of Pavia, Northern Italy. It includes 103 spectral bands with 610×340 pixels, and the spatial resolution is 1.3 m. The dataset contains nine classes and 42 776 labeled pixels.

TABLE I
Number of Training Samples and Test Samples Used for the Indian Pines Dataset

Land cover type	Training labeled samples	Training unlabeled samples	Testing samples
Alfalfa	4	4	38
Corn-notill	142	142	1144
Corn-mintill	83	83	664
Corn	23	23	191
Grass-pasture	48	48	387
Grass-trees	73	73	584
Grass-pasture-mowed	2	2	24
Hay-windrowed	47	47	384
Oats	2	2	16
Soybean-notill	97	97	778
Soybean-mintill	245	245	1965
Soybean-clean	59	59	475
Wheat	20	20	165
Woods	126	126	1013
Buildings-grass-trees-drives	38	38	310
Stone-steel-towers	9	9	75
Total	1018	1018	8213

TABLE II
Number of Training Samples and Test Samples Used for the Salinas Dataset

Land cover type	Training labeled samples	Training unlabeled samples	Testing samples
Weeds-1	40	40	1929
Weeds-2	74	74	3875
Fallow	39	39	1898
Fallow-plow	27	27	1340
Fallow-smooth	53	53	2572
Stubble	79	79	3801
Celery	71	71	3437
Grapes	225	225	10821
Soil	124	124	5955
Corn	65	65	3148
Lettuce-4wk	21	21	1026
Lettuce-5wk	38	38	1851
Lettuce-6wk	18	18	880
Lettuce-7wk	21	21	1028
Vinyard-untrained	145	145	6978
Vinyard-trellis	36	36	1735
Total	1076	1076	51977

First, max-min normalization is used for data preprocessing. For the Salinas dataset and the PaviaU dataset, about 4% of the available samples (including labeled and unlabeled samples) are selected as the training set to train the classifier, and the remaining samples (comprising only unlabeled samples) are used to evaluate the classifier performance. For the Indian Pines dataset, about 20% of the samples are selected as the training set. The detailed information on the training samples and test samples used in our experiments is given in Tables I–III.

B. Comparison With Other Methods

To evaluate the performance of the method presented in this article, we conducted a series of comparisons with the following approaches.

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TABLE III
NUMBER OF TRAINING SAMPLES AND TEST SAMPLES USED FOR THE PAVIAU DATASET

Land cover type	Training labeled samples	Training unlabeled samples	Testing samples
Asphalt	132	132	6367
Meadows	372	372	17905
Gravel	41	41	2017
Trees	61	61	2942
Painted metal sheets	26	26	1293
Bare soil	100	100	4829
Bitumen	26	26	1278
Self-Blocking-Bricks	73	73	3536
Shadows	18	18	911
Total	849	849	41078

1) *Label spreading (LS) + SVM [48], [49]:* In the LS, a classifying function classifies unknown samples by considering the local and global consistency among samples. For the semisupervised comparison experiment, the methods should use the same training set and test set. Here, the SVM classifier and LS are jointly used for the comparison experiment.
2) *Fully connected-ladder network (FC-LN) [28], [50]:* The FC-LN can combine supervised learning with unsupervised learning in deep learning algorithms. The FC-LN minimizes the sum of the supervised and unsupervised cost function, avoiding the need for layerwise pretraining. For the comparison experiments of FC-LN, the encoding block and the decoding block, which, respectively, are consisted of six linear layers, are used for feature extraction. The number of neurons in the last linear layer of the encoding block is consistent with the number of categories in the experimental data. The number of neurons in the last layer of the decoding block is consistent with the feature dimension of the original input data.
3) *Graph convolutional network (GCN) [51]:* The GCN can effectively extract nonlinear spatial–spectral features for semisupervised learning. For the GCN used in the experiments, the graph convolutional hidden layer, including 128 neurons, is used, and the number of neurons in the classification layer depends on the number of classes in the HSI dataset used.
4) *SSCNN [29]:* The SSCNN extracts the spatial–spectral feature from HSI and can simultaneously minimize the sum of the supervised and unsupervised loss function. For the comparison experiments of SSCNN, the supervised block is consisted of a convolutional layer with a filter size of 100, a pooling layer, a BatchNorm (BN) layer, an activator layer, and a linear layer with a Softmax function for classification. The unsupervised block is consisted of a convolutional layer, a pooling layer, four activator layers, four BN layers, and four linear layers.
5) *SAE [25]:* The SAE model can be constructed by stacking several autoencoder blocks and a basic autoencoder model contains an input layer, a hidden layer, and a reconstruction layer.
6) *Semi-GAN (SGAN) [26], [52]:* Semi-GAN extends GAN to the semisupervised classification context by forcing the discriminator network to output class labels. In the comparison experiments, the semi-GAN is consisted of a generator network and a discriminator network. The generator, including three linear layers and three activator layers, can generate fake images of the same size as the input data. The discriminator network, including four linear layers and four activator layers, can simultaneously predict the label distribution of the input data and judge whether the input is a true or false image.

For the LS, the entire training set is used to predict the label information of the unlabeled samples in the training set. Then, the labeled and the unlabeled samples with predicted label information are jointly employed to train a supervised SVM classifier, while the test set is used to evaluate the classification results. For the LS and the trained SVM, the radial basis function is selected as the kernel function, and the model complexity is adjusted by the parameter gamma to obtain the highest classification accuracy. The gamma in the LS is selected from $\{10, 20, 30, \ldots, 100\}$ and is selected from $\{1, 2, 3, \ldots, 10\}$ in the SVM. For the FC-LN, all samples in the training set are used to update the model parameters in a semisupervised manner, and the test samples are fed into the encoding block to obtain the label information of unlabeled samples. For the GCN, the adjacency matrix can be calculated using a k-nearest neighbors (KNN)-based graph and the number of the nearest neighbors is selected from the set $\{3, 4, 5, \ldots, 10\}$. For GCN and SSCNN, only the labeled samples in the training set are employed to update the model parameters, and all test samples are used to test the model performance. For the SAE, the entire training process contains unsupervised pretraining and supervised fine-tuning. In the stage of unsupervised pretraining, all samples in the training set are used to update the model parameters in an unsupervised manner, and the labeled samples are used to fine-tune the SAE model in the supervised fine-tuning stage. For the SGAN, all samples in the training set are employed to train the model parameters of the generator and the discriminator including a Softmax classifier. Then, during the test process, the generator is discarded and the discriminator is used to classify the unlabeled samples.

C. Experimental Results

Three performance metrics, including overall accuracy (OA), average accuracy (AA), and kappa coefficient, are adopted to evaluate the classification results. The classification results on three datasets for all methods are listed in Tables IV–VI and visualized in Figs. 2–4. These results can be summarized as follows.

1) According to the comparison with other methods in Tables IV–VI, the DRFL-VAT achieved better classification performance than other methods. This reveals that our method can effectively utilize the local properties of samples to learn representative features for HSI classification. Moreover, the classification results

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TABLE IV
CLASSIFICATION RESULTS (%) OBTAINED BY SEVERAL METHODS ON THE INDIAN PINES DATASET

Class	LS+SVM	FC-LN	GCN	SSCNN	SAE	SGAN	DRFL-VAT
Alfalfa	65.79	68.42	64.29	2.63	50.00	63.16	84.21
Corn-notill	63.90	83.22	80.79	90.30	73.34	75.96	87.76
Corn-mintill	59.94	82.98	65.60	76.36	61.90	65.36	87.05
Corn	44.50	69.63	62.15	63.35	54.97	61.26	67.02
Grass-pasture	86.30	90.18	92.41	94.06	86.30	89.66	97.93
Grass-trees	92.47	93.49	95.89	99.49	92.64	88.53	98.29
Grass-pasture-mowed	50.00	87.50	73.08	0.00	91.67	83.33	58.33
Hay-windrowed	92.97	97.40	99.07	100.00	98.18	97.92	99.74
Oats	37.50	75.00	38.89	0.00	56.25	50.00	81.25
Soybean-notill	63.50	75.84	76.91	78.92	74.04	75.19	86.50
Soybean-mintill	82.44	84.94	83.57	90.69	80.81	78.42	93.59
Soybean-clean	44.00	80.00	81.84	78.74	70.32	73.89	77.68
Wheat	93.94	96.97	95.68	100.00	96.97	87.27	99.39
Woods	94.97	95.76	95.26	98.22	93.88	93.48	94.57
Buildings-grass-trees-drives	40.65	65.23	58.62	72.26	57.74	58.06	79.35
Stone-steel-towers	77.33	92.00	88.10	88.00	89.33	92.00	100.00
OA	74.42	85.21	83.10	87.80	79.29	79.50	**90.45**
AA	68.14	83.53	78.26	70.81	76.77	77.09	**87.04**
Kappa	0.71	0.83	0.81	0.86	0.76	0.77	**0.89**

The best result are highlight in bold.

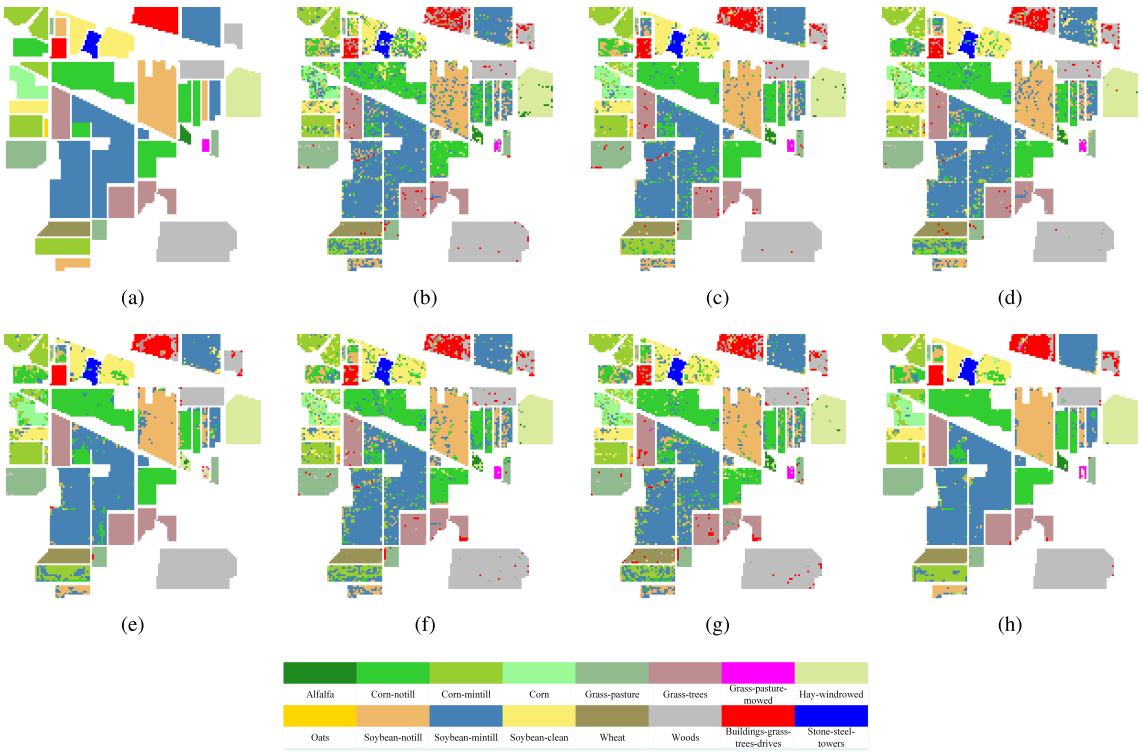


Fig. 2. Classification maps obtained for the Indian Pines dataset by using 10% of the available labeled samples for each class. Ground truth (a) and classification maps obtained by (b) LS+SVM, (c) FC-LN, (d) GCN, (e) SSCNN, (f) SAE, (g) SGAN, and (h) DRFL-VAT.

of DRFL-VAT suggest that the extra loss constraints adopted by our method are effective for improving classification performance.

2) The experimental results of all methods for the Indian Pines dataset are generally worse than those obtained with the two remaining datasets. The possible reasons are the remaining datasets have higher spatial resolution and the class imbalance phenomenon.

D. Validity Analysis for the LML and VAT

In Fig. 5, the proposed DRFL-VAT method is compared with other variants of the CNN model, including the basic CNN, the CNN-LML (which is obtained by embedding the LML into each fully connected layer of the basic CNN), and the CNN-VAT (which integrates the VAT regularization into the basic CNN). The structure of basic CNN is the same as the CNN structure shown in Table VII. Compared

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TABLE V
CLASSIFICATION RESULTS (%) OBTAINED BY SEVERAL METHODS ON THE SALINAS DATASET

Class	LS+SVM	FC-LN	GCN	SSCNN	SAE	SGAN	DRFL-VAT
Weeds-1	96.94	99.95	97.26	98.19	98.16	97.41	99.84
Weeds-2	99.19	99.25	99.78	99.89	99.79	99.39	98.97
Fallow	98.52	99.63	95.97	99.89	97.41	98.31	99.68
Fallow-plow	98.96	99.03	99.05	98.36	99.52	98.88	99.78
Fallow-smooth	97.90	97.40	98.97	98.21	99.05	98.37	98.25
Stubble	99.79	99.76	99.85	99.97	99.86	99.37	99.58
Celery	99.77	99.62	99.86	98.94	99.38	98.31	99.65
Grapes	83.22	80.69	81.66	86.97	81.62	80.55	83.01
Soil	99.82	99.82	99.92	99.92	99.57	99.28	99.68
Corn	91.80	95.68	92.56	92.56	92.27	91.39	94.50
Lettuce-4wk	96.00	96.26	97.80	97.56	92.93	97.86	98.34
Lettuce-5wk	99.68	99.73	99.95	100.00	99.60	98.81	100.00
Lettuce-6wk	97.50	98.20	98.00	100.00	99.39	95.57	99.66
Lettuce-7wk	96.01	96.40	96.57	97.28	93.67	89.30	97.18
Vinyard-untrained	63.94	74.23	69.44	73.80	68.07	68.44	82.43
Vinyard-trellis	96.54	95.68	98.48	98.33	98.46	94.81	97.23
OA	90.44	91.46	91.06	93.10	89.50	90.10	**93.31**
AA	94.72	95.14	95.32	96.51	93.83	94.13	**96.74**
Kappa	0.89	0.91	0.90	0.92	0.88	0.89	**0.93**

The best result are highlight in bold.

TABLE VI
CLASSIFICATION RESULTS (%) OBTAINED BY SEVERAL METHODS ON THE PAVIAU DATASET

Class	LS+SVM	FC-LN	GCN	SSCNN	SAE	SGAN	DRFL-VAT
Asphalt	89.34	91.13	93.08	94.91	89.46	90.67	95.99
Meadows	98.35	97.68	96.62	98.58	96.59	97.02	98.76
Gravel	64.25	67.03	68.61	82.50	66.93	70.45	83.79
Trees	88.21	93.61	93.31	89.84	87.56	88.92	94.43
Painted metal sheets	98.61	99.77	97.88	98.45	98.92	98.76	98.22
Bare soil	73.02	82.05	81.03	90.49	78.55	80.97	88.59
Bitumen	76.76	90.20	76.69	61.03	86.70	85.45	80.83
Self-Blocking-Bricks	83.03	86.03	85.48	84.59	85.92	83.43	88.43
Shadows	99.67	99.56	96.99	96.82	99.89	99.56	95.83
OA	89.62	91.90	91.09	93.23	90.18	90.84	**94.56**
AA	85.69	89.67	87.74	88.58	87.83	88.36	**91.65**
Kappa	0.86	0.89	0.88	0.91	0.87	0.88	**0.93**

The best result are highlight in bold.

TABLE VII
CONFIGURATION OF THE CNN ARCHITECTURE IN OUR METHOD FOR THE THREE DATASETS

Convolutional layer and pooling layer				Fully connected layers	
Convolutional kernel		Function	Pooling kernel	The number of neuron	Function
Multiscale filter bank	$256 \times 5 \times 5 \times d$	LeakyReLU	2×2	128	LeakyReLU
	$256 \times 3 \times 3 \times d$			32	LeakyReLU
	$256 \times 1 \times 1 \times d$			c	Softmax
$256 \times 3 \times 3 \times 768$		LeakyReLU	2×2		

with the basic CNN, improvements in terms of OA can be observed in the CNN-LML and CNN-VAT, which reveals that the LML and VAT regularization are effective for enhancing the classification performance. Compared with other CNN methods in Fig. 5, the proposed DRFL-VAT obtained the best classification results.

To better understand the effectiveness of our proposed model, we used t-distributed stochastic neighbor embedding (t-SNE) [53] to reduce the dimensionality of the features of the test samples, which include the original data, the last hidden layer output features extracted by the basic CNN and DRFL-VAT, to two dimensions. The 2-D feature representations are visualized and listed in Fig. 6, with different colors representing different classes in the HSI dataset. As shown in Fig. 6, the deep features extracted by the basic CNN and DRFL-VAT are more discriminative than the original HSI, which means that the separability between the feature representations of different classes is improved. Moreover, due to the introduction of LML loss and VAT loss, the features extracted by DRFL-VAT have slightly improved the separability between different classes compared to those extracted by the basic CNN model.

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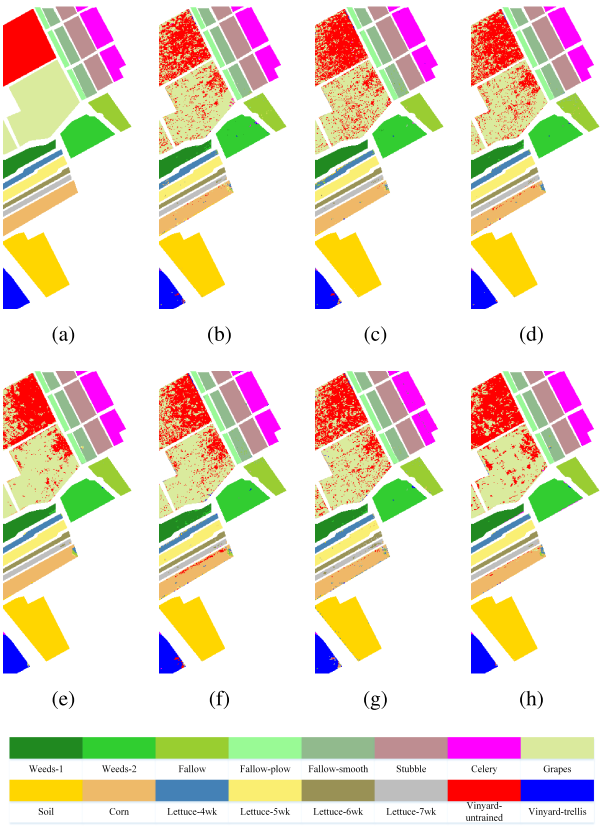


Fig. 3. Classification maps obtained for the Salinas dataset by using 2% of the available labeled samples for each class. Ground truth (a) and classification maps obtained by (b) LS+SVM, (c) FC-LN, (d) GCN, (e) SSCNN, (f) SAE, (g) SGAN, and (h) DRFL-VAT.

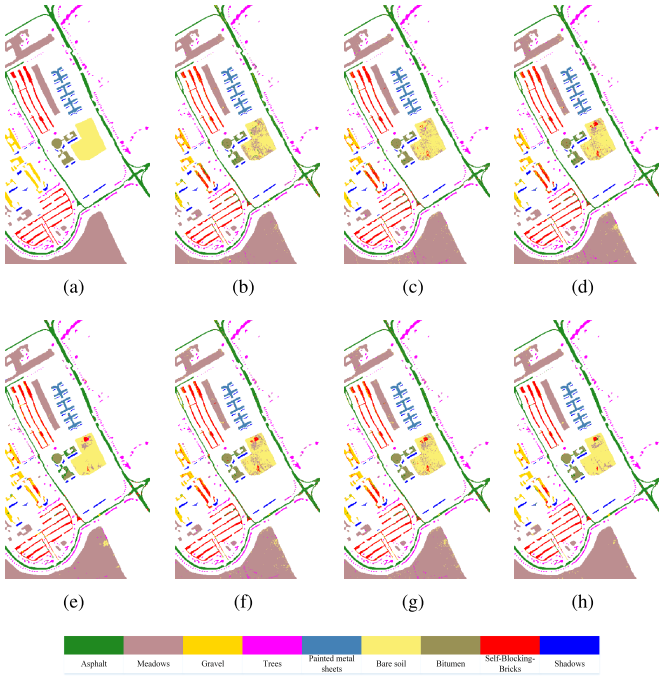


Fig. 4. Classification maps obtained for the PaviaU dataset by using 2% of the available labeled samples for each class. Ground truth (a) and classification maps obtained by (b) LS+SVM, (c) FC-LN, (d) GCN, (e) SSCNN, (f) SAE, (g) SGAN, and (h) DRFL-VAT.

E. Parameter Analysis

In our DRFL-VAT, the parameters need to be tuned for each dataset. The parameters can be divided into two types: the model architecture parameters and the hyperparameters, which include λ_1, λ_2, ε, ξ, and ω. For the model architecture parameters, the detailed parameter settings are listed in Table VII. c is the number of categories of the HSI dataset used, and d is the number of bands. The negative slope of the LeakyReLU function is set to 0.1 in our experiments. For the hyperparameters, we set $\xi = 0.01$ and $\omega = 0.001$ and the influence of the parameters λ_1, λ_2, and ε are evaluated on the HSI classification results. Under different parameter conditions, the OA of three experimental datasets is shown in Fig. 7. The parameters which provide the best classification results are different for each dataset, since each dataset has its own distinctive land cover distribution. For the Salinas dataset, the classification

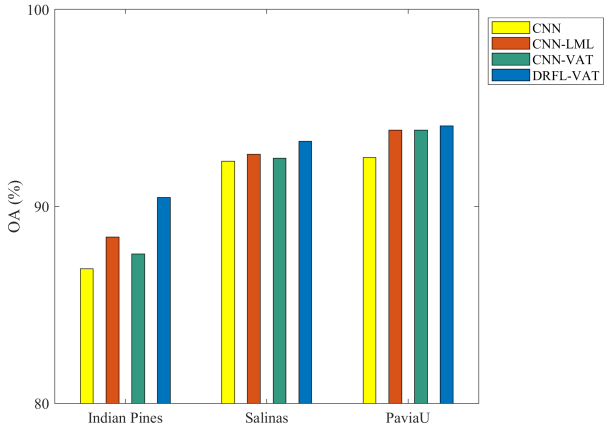


Fig. 5. Analyzing the importance of the LML and VAT constraint.

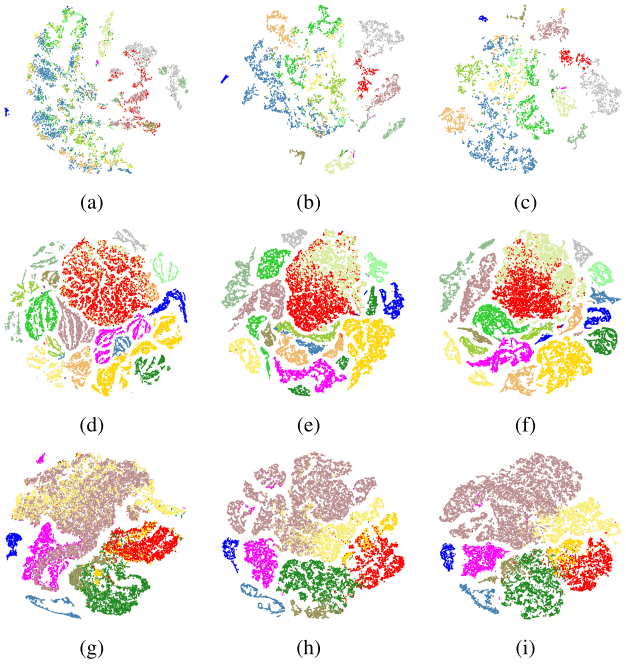


Fig. 6. Analysis for separability of feature representation. Two-dimensional embedding visualizations on the Indian Pines dataset for (a) original data, (b) features extracted by the basic CNN, and (c) features extracted by the DRFL-VAT. The visualizations on the Salinas dataset for (d) original data, (e) features extracted by the basic CNN, and (f) features extracted by the DRFL-VAT. The visualizations on the PaviaU dataset for (g) original data, (h) features extracted by the basic CNN, and (i) features extracted by the DRFL-VAT.

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TABLE VIII
CLASSIFICATION RESULTS ON DIFFERENT CONVOLUTIONAL KERNEL COMBINATION OF MULTISCALE FILTER BANK IN DRFL-VAT

Dataset		The structure of multi-scale filter bank					
		1×1, 3×3	5×5	1×1, 5×5	1×1, 3×3, 5×5	1×1, 3×3, 7×7	1×1, 5×5, 7×7
Indian pines	OA(%)	86.59	87.28	88.04	90.45	90.33	90.02
	AA(%)	79.25	78.36	81.24	87.04	81.55	83.29
	Kappa	0.85	0.85	0.86	0.89	0.89	0.89
Salinas	OA(%)	92.25	92.29	92.93	93.31	93.25	93.65
	AA(%)	96.07	95.26	95.89	96.74	96.21	96.48
	Kappa	0.91	0.91	0.92	0.93	0.92	0.93
PaviaU	OA(%)	93.73	94.05	94.07	94.56	93.88	94.35
	AA(%)	91.31	90.52	90.83	91.65	90.45	90.71
	Kappa	0.92	0.92	0.92	0.93	0.92	0.92

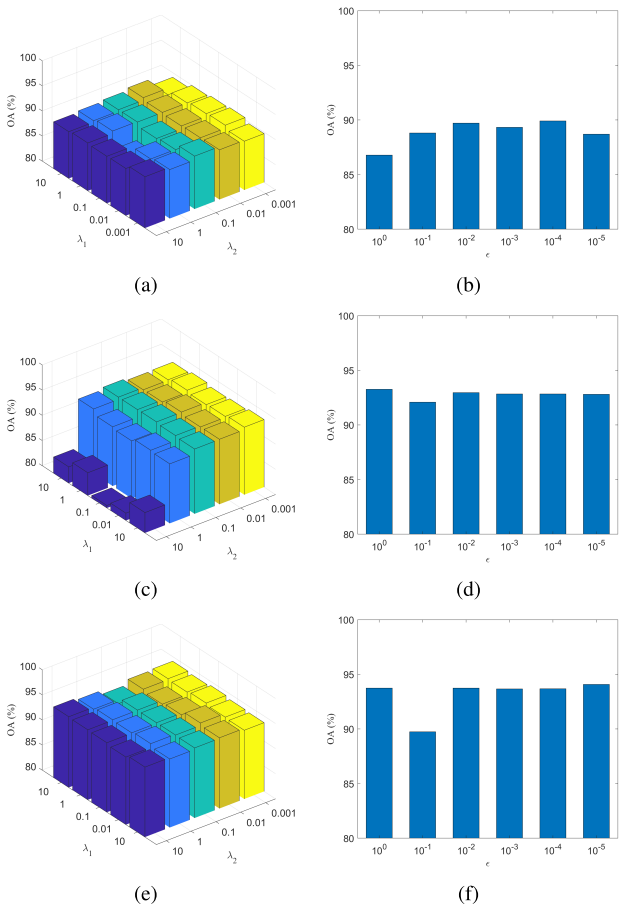


Fig. 7. Influence of the different values of the parameters on the three considered datasets. (a) and (b) Impact of parameters λ_1, λ_2, and ε on the classification performance for the Indian Pines dataset. (c) and (d) Impact of parameters λ_1, λ_2, and ε on the classification performance for the Salinas dataset. (e) and (f) Impact of parameters λ_1, λ_2, and ε on the classification performance for the PaviaU dataset. In (a), (c), and (e), ε is set to 10^{-4}, 1, and 10^{-5}, respectively. In (b), (d), and (f), we set $\lambda_1 = 1$ and $\lambda_2 = 0.1$.

performance is poor when the value of parameter λ_2 is larger. One possible reason is that it might be influenced by the adversarial noise in the VAT loss. For the three experimental datasets, a lower value of ε can generally lead to stable performance.

It is worth noting that a multiscale filter bank is set at the beginning of the CNN to extract both spatial and spectral information in HSI. To analyze the effectiveness of the multiscale filter bank, we conducted a series of classification experiments for different convolutional kernel combinations. In the experiments, model architecture parameters and the input data size for our model are fine-tuned for different convolutional kernel combinations. The results of convolutional kernel 5×5 are chosen as the baseline and the results of 1×1, 5×5 verify the validity of the combination of spectral information and spatial information. Meanwhile, the experiment results, which are listed in columns 3–5 of Table VIII, show that moderate choosing a larger kernel size is beneficial to the improvement of classification performance. Moreover, we analyze the effect of increasing the filter size in the multiscale filter bank on classification accuracy. A series of experiments, which are listed in columns 7–8 of Table VIII, are conducted by choosing 7×7 as the largest convolutional kernel size. Compared with the results of 5×5 as the largest convolutional kernel size, the results of 7×7 do not improve significantly and it increases the network parameters and computation cost. Finally, the results of the combination of 1×1, 3×3 and 5×5 demonstrate the effectiveness of the structures used in Table VII.

IV. CONCLUSION AND FUTURE WORK

In this article, we developed a novel semisupervised classification framework based on CNN for HSI classification purposes. Semisupervised deep learning methods are very effective in remote-sensing applications because they can overcome the limited availability of labeled samples (which are very difficult to obtain in practice). Taking advantage of the VAT loss and LML loss, the supervised CNN model is extended to semisupervised learning problems by utilizing both labeled and unlabeled samples. Both the geometry manifold and the local information among samples are explored to obtain more representative features for HSI classification purposes. Our experimental results with three public datasets demonstrate the effectiveness of the proposed method. In the future, we will explore better ways to utilize unlabeled samples

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for classification purposes by combining our method with SAE or GAN.

APPENDIX

In this section, the gradient of LML loss Φ_{LML} in the fully connected layers is summarized. The derivative of LML loss with respect to the parameters in each fully connected layer is computed. Using (1), (2), (5), and (6), the gradient of LML loss with respect to $\mathbf{W}^{(u)}$ and $\mathbf{B}^{(u)}$ is given by

$$\begin{cases} \dfrac{\partial \Phi_{\mathrm{LML}}}{\partial \mathbf{W}^{(u)}} = \displaystyle\sum_{k=1}^{M} \dfrac{\partial \Phi_{\mathrm{LML}}}{\partial \mathbf{P}_{\cdot k}^{(u)}} \left(\mathbf{Z}_{\cdot k}^{(u-1)}\right)^T \\ \dfrac{\partial \Phi_{\mathrm{LML}}}{\partial \mathbf{B}^{(u)}} = \displaystyle\sum_{k=1}^{M} \dfrac{\partial \Phi_{\mathrm{LML}}}{\partial \mathbf{P}_{\cdot k}^{(u)}} \end{cases} \tag{13}$$

where $u = 1, 2, 3$. For (13), the following result needs to be computed:

$$\begin{cases} \dfrac{\partial \Phi_{\mathrm{LML}}}{\partial \mathbf{P}_{\cdot k}^{(3)}} = 2 \displaystyle\sum_{n=1}^{M} \left(\left(\mathbf{V}_{\mathrm{kn}}^{(3)} - \dfrac{\mathbf{V}_{\mathrm{kn}}^{(3)}}{\omega} \left\| \mathbf{Z}_{\cdot k}^{(3)} - \mathbf{Z}_{\cdot n}^{(3)} \right\|_2^2 \right) \right. \\ \qquad \left. \otimes \left(\mathbf{Z}_{\cdot k}^{(3)} - \mathbf{Z}_{\cdot n}^{(3)} \right) \right) \odot \dfrac{\partial \mathbf{Z}_{\cdot k}^{(3)}}{\partial \mathbf{P}_{\cdot k}^{(3)}} \\ \dfrac{\partial \mathbf{Z}_{\mathrm{ik}}^{(3)}}{\partial \mathbf{P}_{\mathrm{ik}}^{(3)}} = 1 \end{cases} \tag{14}$$

where $\otimes$ represents the Kronecker product and $\odot$ represents the Hadamard product. Meanwhile, we have the result as follows:

$$\begin{aligned} \frac{\partial \Phi_{\mathrm{LML}}}{\partial \mathbf{P}_{\cdot k}^{(t)}} = & \left((\mathbf{W}^{(t+1)})^T \frac{\partial \Phi_{\mathrm{LML}}}{\partial \mathbf{P}_{\cdot k}^{(t+1)}} \right) \odot \frac{\partial \mathbf{Z}_{\cdot k}^{(t)}}{\partial \mathbf{P}_{\cdot k}^{(t)}} \\ & + 2 \sum_{n=1}^{M} \left(\left(\mathbf{V}_{kn}^{(t)} - \frac{\mathbf{V}_{kn}^{(t)}}{\omega} \left\| \mathbf{Z}_{\cdot k}^{(t)} - \mathbf{Z}_{\cdot n}^{(t)} \right\|_2^2 \right) \right. \\ & \left. \otimes \left(\mathbf{Z}_{\cdot k}^{(t)} - \mathbf{Z}_{\cdot n}^{(t)} \right) \right) \odot \frac{\partial \mathbf{Z}_{\cdot k}^{(t)}}{\partial \mathbf{P}_{\cdot k}^{(t)}} \end{aligned} \tag{15}$$

where $t = 1, 2$. When the LeakyReLU function is chosen as the activation function, we have the following result:

$$\frac{\partial \mathbf{Z}_{ik}^{(t)}}{\partial \mathbf{P}_{ik}^{(t)}} = \begin{cases} 1, & \text{if } \mathbf{P}_{ik}^{(t)} \geq 0 \\ \alpha, & \text{otherwise} \end{cases} \tag{16}$$

where α is the negative slope of the LeakyReLU function. In addition, to calculate the gradient of LML in the layers before the fully connected layers by the existing open source framework, the gradient of LML loss with respect to the input data $\mathbf{Z}^{(0)}$ of the first fully connected layer is given by

$$\frac{\partial \Phi_{\mathrm{LML}}}{\partial \mathbf{Z}_{\cdot k}^{(0)}} = (\mathbf{W}^{(1)})^T \frac{\partial \Phi_{\mathrm{LML}}}{\partial \mathbf{P}_{\cdot k}^{(1)}}. \tag{17}$$

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